Advanced Packaging
A New Growth Engine for KLA

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KLA Corporation

Nasdaq Jefferies

June 11, 2024

Forward Looking Statements

Statements in this presentation other than historical facts, such as statements pertaining to: (i) future industry demand for semiconductors and WFE; (ii) our market position for the future and timing for the resumption of growth in demand for our products; and (iii) our revenues from advanced packaging for 2024 and beyond, are forward-looking statements and subject to the Safe Harbor provisions created by the Private Securities Litigation Reform Act of 1995.

These forward-looking statements are based on current information and expectations and involve a number of risks and uncertainties. Actual results may differ materially from those projected in such statements due to various factors, including but not limited to: our vulnerability to a weakening in the condition of the financial markets and the global economy; risks related to our international operations; evolving Bureau of Industry and Security of the U.S. Department of Commerce rules and regulations and their impact on our ability to sell products to and provide services to certain customers in China; costly intellectual property disputes that could result in our inability to sell or use the challenged technology; risks related to the legal, regulatory and tax environments in which we conduct our business; increasing attention to ESG matters and the resulting costs, risks and impact on our business; unexpected delays, difficulties and expenses in executing against our environmental, climate, diversity and inclusion or other ESG target, goals and commitments; our ability to attract, retain and motivate key personnel; our vulnerability to disruptions and delays at our third party service providers; cybersecurity threats, cyber incidents affecting our and our business partners' systems and networks; our inability to access critical information in a timely manner due to system failures; our ability to identify suitable acquisition targets and successfully integrate and manage acquired businesses; climate change, earthquake, flood or other natural catastrophic events, public health crises such as the COVID-19 pandemic or terrorism and the adverse impact on our business operations; the wars between Israel and Hamas and Russia and Ukraine, and the significant military activity in that region; lack of insurance for losses and interruptions caused by terrorists and acts of war, and our self-insurance of certain risks including earthquake risk; risks related to fluctuations in foreign currency exchange rates; risks related to fluctuations in interest rates and the market values of our portfolio investments; risks related to tax and regulatory compliance audits; any change in taxation rules or practices and our effective tax rate; compliance costs with federal securities laws, rules, regulations, NASDAQ requirements, and evolving accounting standards and practices; ongoing changes in the technology industry, and the semiconductor industry in particular, including future growth rates, pricing trends in end-markets, or changes in customer capital spending patterns; our vulnerability to a highly concentrated customer base; the cyclicality of the industries in which we operate; our ability to timely develop new technologies and products that successfully address changes in the industry; our ability to maintain our technology advantage and protect proprietary rights; our ability to compete in the industry; availability and cost of the materials and parts used in the production of our products; our ability to operate our business in accordance with our business plan; risks related to our debt and leveraged capital structure; we may not be able to declare cash dividends at all or in any particular amount; liability to our customers under indemnification provisions if our products fail to operate properly or contain defects or our customers are sued by third parties due to our products; our government funding for R&D is subject to audit, and potential termination or penalties; we may incur significant restructuring charges or other asset impairment charges or inventory write offs; and risks related to receivables factoring arrangements and compliance risk of certain settlement agreements with the government. For other factors that may cause actual results to differ materially from those projected and anticipated in forward-looking statements in this press release, please refer to KLA's Annual Report on Form 10-K for the year ended June 30, 2023, and other subsequent filings with the Securities and Exchange Commission (including, but not limited to, the risk factors described therein). KLA assumes no obligation to, and does not currently intend to, update these forward-looking statements.

KLA is a Diversified Leader and Critical Technology Enabler

KLA at a Glance[1] (NASDAQ: KLAC)

Global Technology Leader

FORTUNE World's Most Admired Companies

TIME World's Best Companies 2023

FORTUNE 500 2023

~14,700
Employees

>65%
PhD/Master's among professional roles

Global Market Share
>4x
Nearest Competitor [2]

$9.7B
CY23 Revenue
-8% Y/Y Growth

61.7% / 39.5%
Industry Leading Gross/Operating Margins

$2B / >75% under contract
Services Revenue

$22.77 / 27%
CY23 EPS CAGR (19-23)

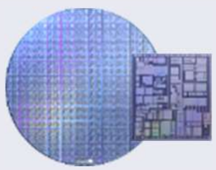
$3.2B / 33%
CY23 FCF / CY23 FCF Margin

[1] As of 12/31/23
[2] Gartner April 2024

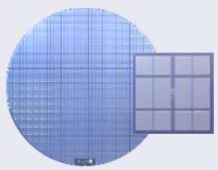
KLA

KLA's Role in the Electronics Ecosystem

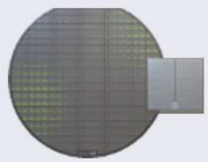
CHIPS



Logic | Foundry

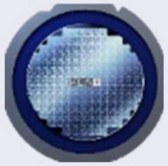


Memory

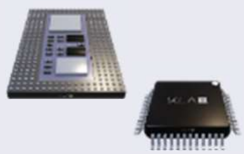


Specialty | Legacy

WAFER-LEVEL PACKAGING

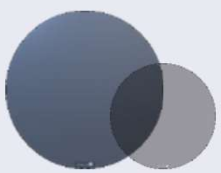


COMPONENTS

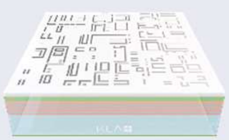


PRINTED CIRCUIT BOARDS

WAFERS

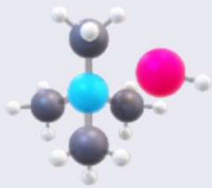


RETICLES

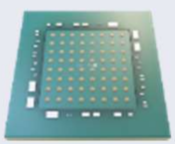


SEMI OEMs

CHEMICALS | MATERIALS



IC SUBSTRATES



END MARKETS

- Data Center
- Artificial Intelligence
- 5G Infrastructure
- Mobile
- Automotive
- PC / Tablet
- Metaverse (AR/VR)

KLA Organizational Structure

Rick Wallace

President and Chief Executive Officer

KLA

Electronics, Packaging & Components (EPC)

Oreste Donzella

Executive Vice President, Electronics, Packaging and Components (EPC)

Semiconductor Process Control

Ahmad Khan

President Semiconductor Process Control (Semi PC)

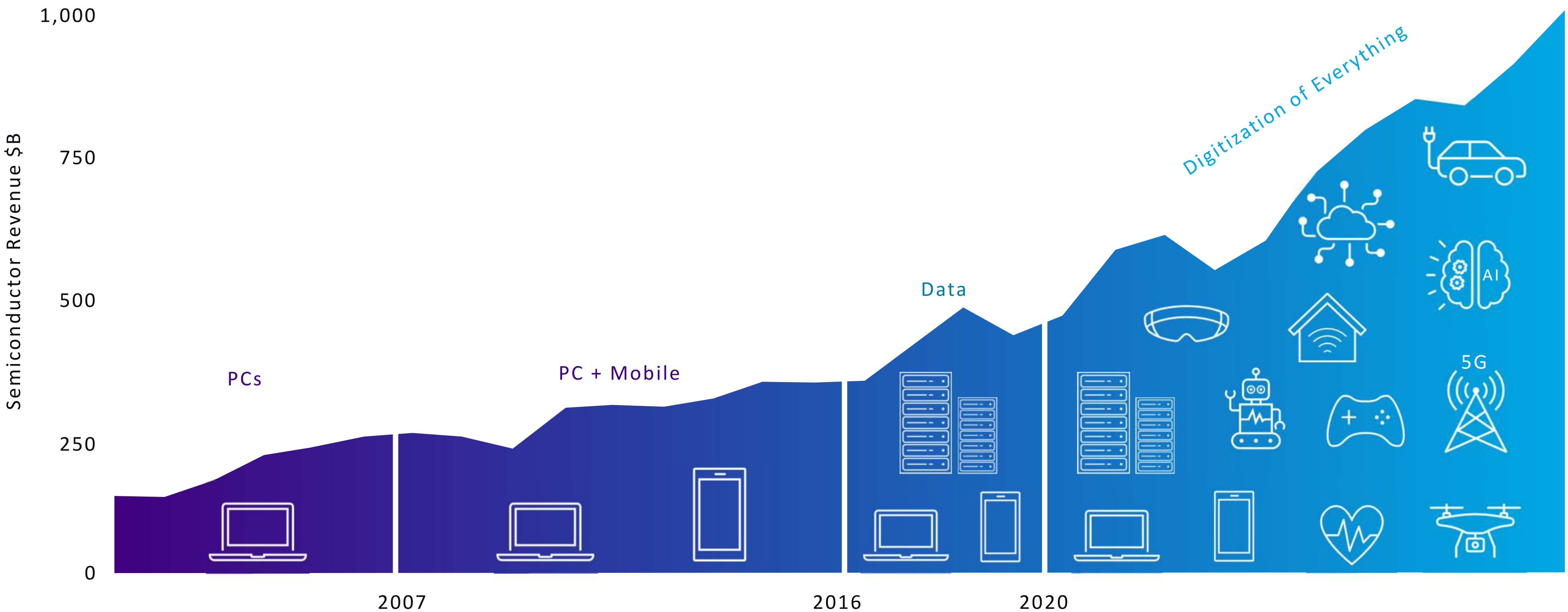
KLA Services

Brian Lorig

Executive Vice President, KLA Services

Semiconductor Key Market Drivers

Semiconductor End Demand has Become More Diversified



Sources: KLA Analysis, TechInsights Sept 2023

AI: The New Driver of the Semiconductor Industry

M40

GPU Package

12 x 1 GB
GDDR5 chips

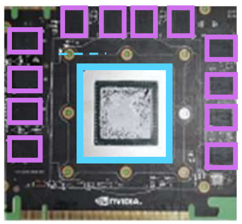


Image source: Nvidia

B100

2 reticle-sized chips

192 GB: 8x24 GB
HBM3e chips

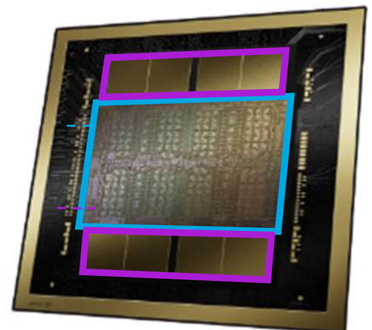
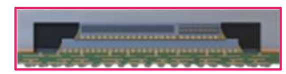


Image source: Nvidia

CoWoS package

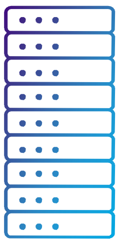


	K40	M40	P100	V100	A100	H100	B100
Year	2013	2015	2016	2017	2020	2022	2024
Area (mm²)	551	601	610	815	826	814	~2x800
Transistors	7.1 B	8.0 B	15.3 B	21.1 B	54.2 B	80 B	208 B
Node	N28	N28	N16	N12	N7	N4	N4P
Memory Type	GDDR5	GDDR5	HBM2	HBM2	HBM2e	HBM3	HBM3e
Memory Size	12 GB	12 GB	16 GB	40 GB	60 GB	80 GB	192 GB
Package Type	FCBGA	FCBGA	CoWoS	CoWoS	CoWoS	CoWoS	CoWoS

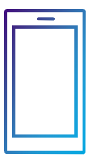
Source: KLA Analysis

AI: Driving Industry Momentum | Semiconductor Content

AI Drives End Markets



HPC



Mobile

PC

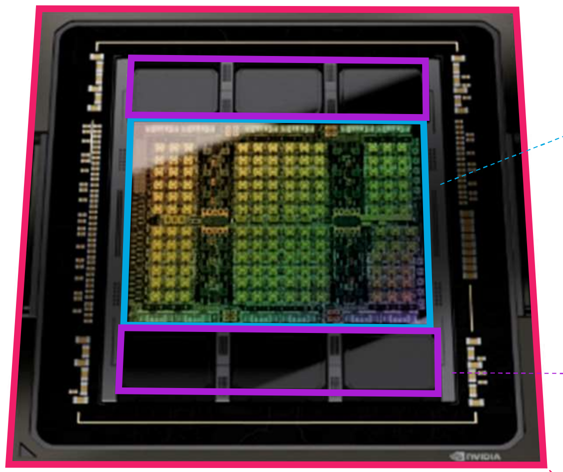
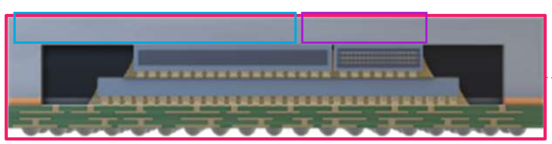


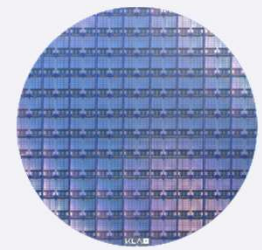
Image source: Nvidia H100



AI Requires Advanced Process Control

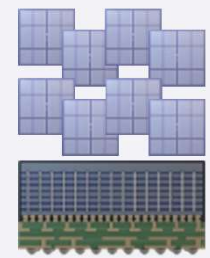
GPU

- Leading-edge logic: ≤3nm design nodes and advanced transistor architectures
- Large die size: ≥800mm²
- Increased design starts



Memory

- Multiple HBM chips / GPU
- ~2.2x more memory vs regular DRAM
- Increased design starts



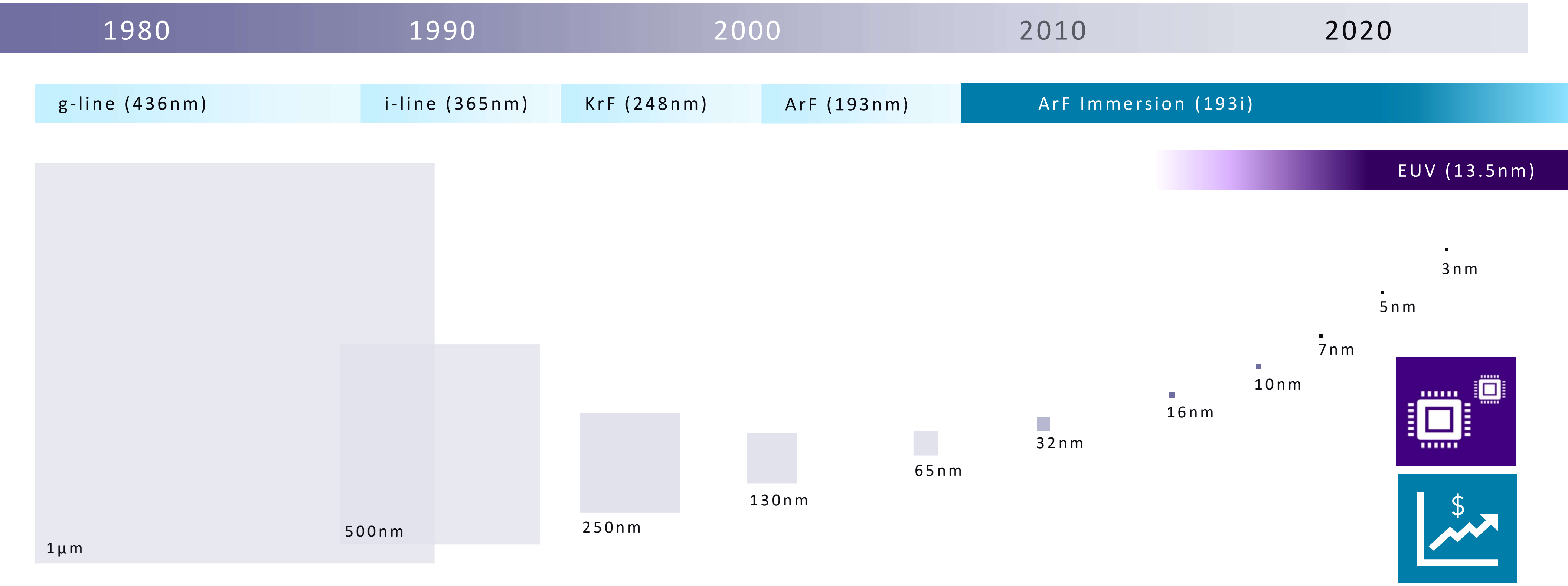
Advanced Packaging

- Large, complex, expensive package
- High semiconductor content

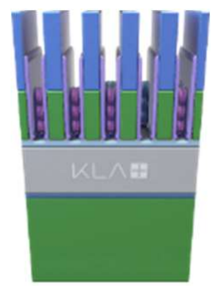
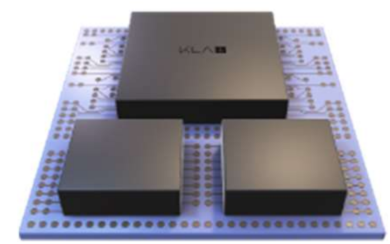
HBM: high bandwidth memory

Semiconductor Technology Roadmap

Semiconductor Technology has Been Scaling for 50+ Years



Scaling is Now Evolving Through Diversified Technology Innovations

Lithography Scaling	Transistor Architecture	Power Distribution Designs	Packaging Heterogeneous Integration
			

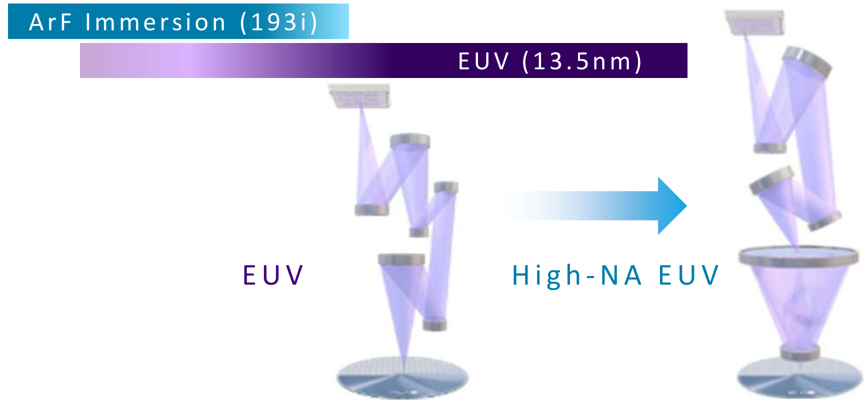
Traditional areas of “Moore’s Law” innovation

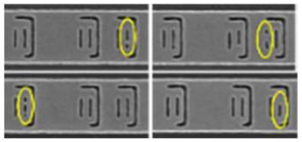
New areas of innovation

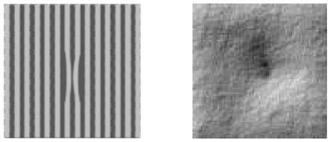
Transition to High-NA EUV Requires Tighter Qualification and Control

Key advantages for high-NA EUV

- Even smaller features and higher density
- Increased pattern uniformity
- Improved productivity







Reticle Qualification Challenges	Process Qualification Challenges	Process/Line Monitoring Challenges
▪ Mask shop qualification for patterned EUV reticles ▪ Reticle re-qual in the IC fab	▪ Process window discovery: stochastics, hotspot prediction ▪ Thin photoresists ▪ EUV materials qualification	▪ Process random & systematic defects ▪ Chamber cleanliness control driven by vacuum environment

Shift to GAA Brings Defectivity and Variability Challenges

Key advantages for GAA (Gate-All-Around)

- Lower leakage current
- Higher transistor density
- Increased effective channel area
- Reduce transistor variability

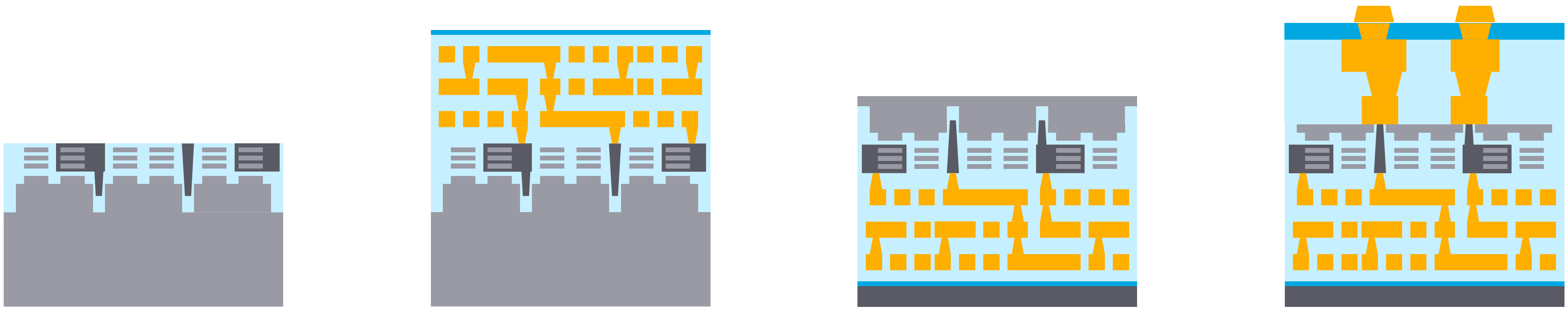
FinFET

GAA

	FinFET and GAA	GAA Unique
Inspection to control defectivity	▪ Mask variation / complexity ▪ Contact module integration ▪ Multi-patterning hotspots ▪ Channel mobility	▪ Si / SiGe super lattice ▪ Nanosheet release ▪ Channel release residues ▪ S/D voids
Metrology to control variability	▪ Lithography simulations ▪ Scatterometry: CD & shape ▪ Optical overlay ▪ Film thickness	▪ Stress ▪ Film composition ▪ SEM overlay

Backside Distribution Network Saves Space and Increases Performance

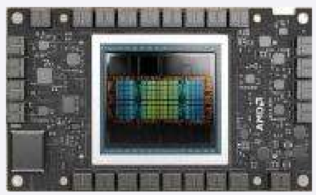
Power via formation in FEOL	Frontside BEOL (conventional)	Face-down wafer bonding then thinning	Via reveal, backside BEOL and bumping



Challenges	Challenges	Challenges	Challenges
Power via alignment	Bonding film quality	Low-distortion wafer bonding	TSV reveal process/metrology
CD and shape		Uniform grinding/thinning	Backside feature alignment
High aspect ratio defectivity		Defectivity	Bumping

Advanced Packaging Shifting to Heterogeneous Integration

Monolithic chip

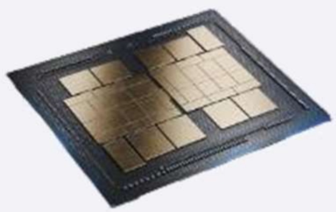


AMD Instinct™ MI300
GPU Accelerators

Source: AMD

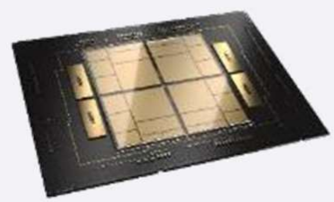
NVidia H100 Tensor
Core GPU

Source: NVidia



Intel® Data Center
GPU Max Series

Source: Intel



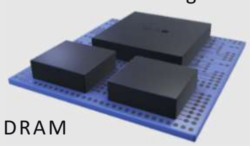
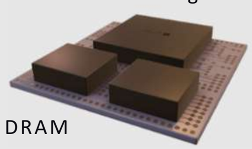
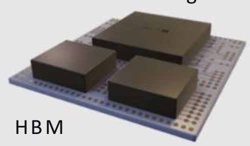
Intel Xeon CPU
Max Series

Source: Intel

Key advantages for
heterogeneous integration

- Higher performance
- Lower latency
- Lower power
- Lower cost

Heterogenous Integration at the Package Level

Low Bandwidth	Wide I/O			High Bandwidth	Ultra-High Bandwidth	
2D Fan-out RDL (Wafer or Panel)			2.5D (Interposer or Bridge)	3D Stacking (µBump or Hybrid Bonding)		
CPU/GPU Logic  DRAM (IC Substrate)	*Mobile* LPDDR DRAM  (POP) Logic	Logic  DRAM (RDL)	*CPU/GPU* Logic  HBM (Silicon)	Buffer-less HBM  Logic	*ASIC* DRAM Cache  Logic	*CPU/GPU* SRAM Cache  Logic
High Volume HVM	High Volume HVM	Med Vol HVM	Med Vol HVM	Development	Low Volume HVM	Low Volume HVM
< 60 GB/s		60 – 196 GB/s	196 GB/s – 1 TB/s			> 1 TB/s

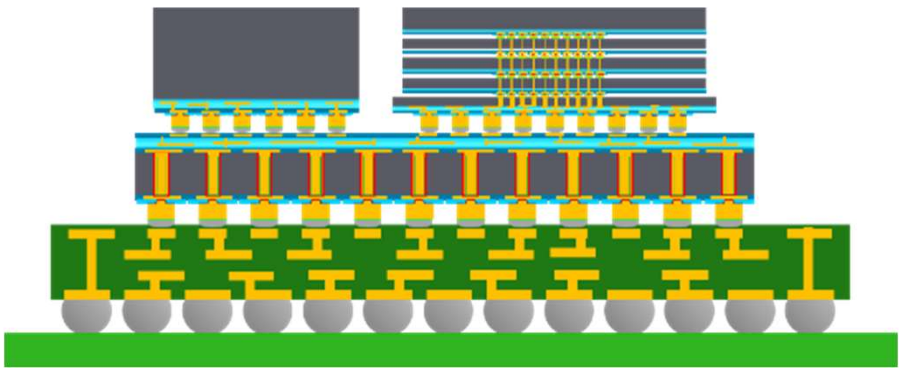
Lower Cost

Higher Performance

Source: Internal KLA Analysis

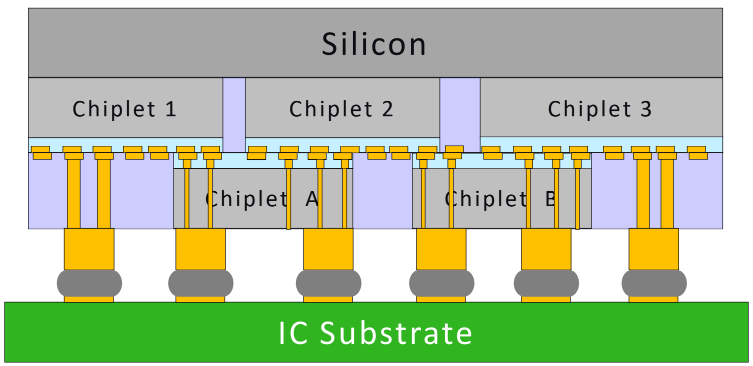
Significant proliferation of packaging architectures depending on end market applications

Key Trends in 2.5D Packaging (e.g., CoWoS)



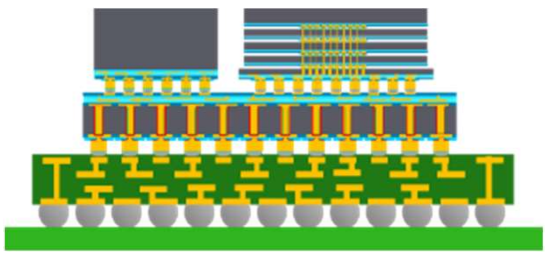
Lateral Interconnect Line/Space	Silicon interposer	0.4µm → 0.2µm
	RDL	2µm
Vertical Interconnect Pitches	Die to interposer	25µm → 15µm
	Interposer to substrate	90µm → 70µm
Package	Interposer size	45mm → 90mm
	Interposer form factor	Wafer → Panel
	Body Size	55mm → 100mm

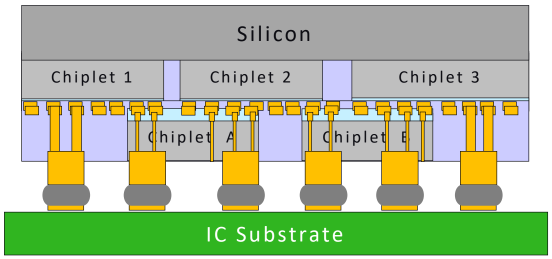
Key Trends in 3D Packaging (e.g., Foveros Direct, SoIC)



Bonding Interconnects	D2W hybrid bonding	9µm → 3µm
	µBump	25µm → 10-15µm
Vertical Interconnects	TSV	9µm → 3µm
	TDV	100µm → 50µm
Post-Bonding Processes	Gap fill	20-40µm → 70-80µm
	Die/wafer thinning	30-40µm → 10-20µm
	Multi-die stack	2 → 4 (Logic) 8/12 → 16 (Memory)

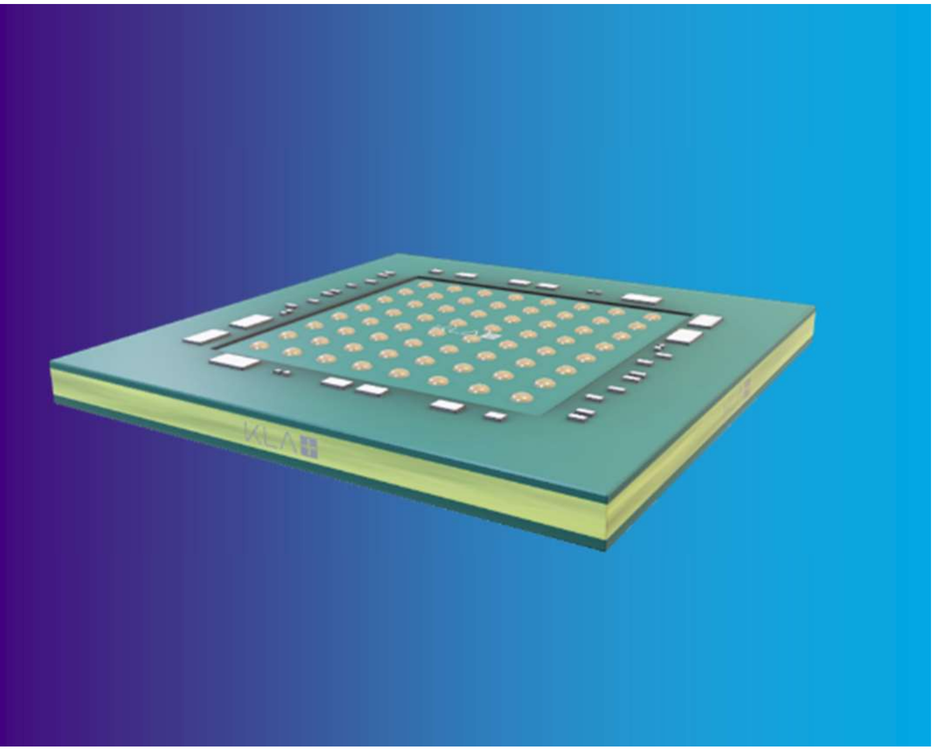
2.5D and 3D Packaging Challenges





	2.5D Challenges	3D Challenges
Process	▪ Wafer thinning ▪ UBM contact resistance ▪ RDL barrier seed quality ▪ TSV formation, Via reveal	▪ Wafer and die thinning ▪ Oxide gap fill ▪ Bonding film strength and uniformity ▪ Die singulation
Inspection	▪ RDL surface defectivity	▪ Bonding surface cleanliness
Metrology	▪ TSV profile ▪ RDL CD and height ▪ Plating chemistry	▪ Bonding surface profile ▪ Wafer and die warpage ▪ Chiplet bonding alignment

IC Substrates Roadmap is Essential to Heterogenous Integration

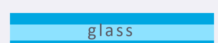


Shrinking feature sizes

Increase in number of layers

Larger package size

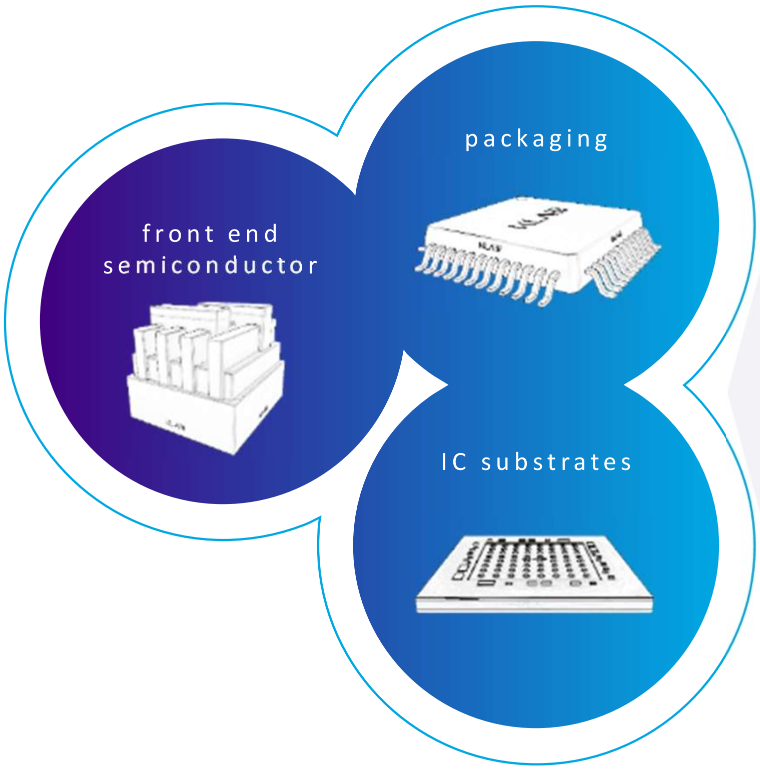
Thinner laminates



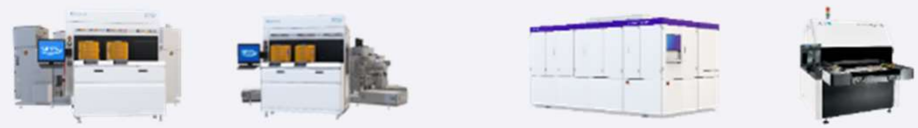
New materials

KLA in Packaging

Differentiated and Diversified Product Portfolio



Process



plasma etch, plasma dicing, PECVD and PVD for wafers

direct imaging, automated optical shaping for IC substrates

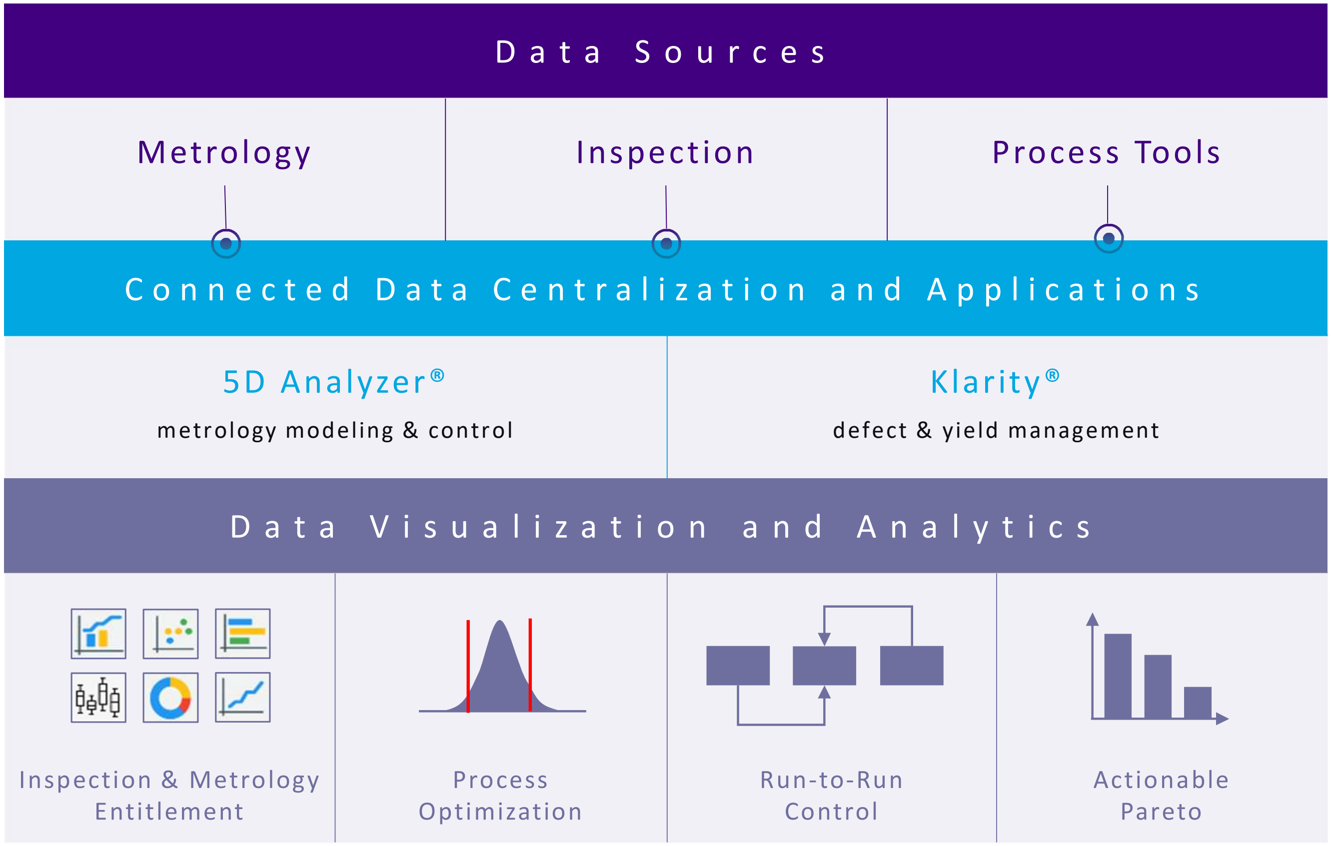
Inspection

inspection solutions and analytics for wafer-level packaging, component assembly/test and IC substrates

Metrology

metrology solutions and analytics for wafer (shape, films, overlay, CD), component (3D) and IC substrates (CD, films, topography)

Data Automation and Analytics Solutions

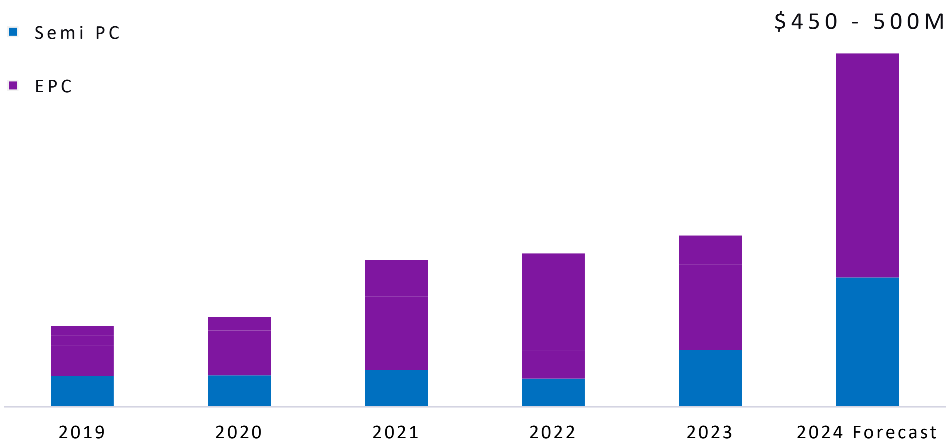


Bringing data analytical capabilities from front end to 3D packaging is critical for yield

Advanced Packaging: A New Growth Engine for KLA

~25%	~5x	>50	>50
KLA Packaging System Revenue CAGR[1]	KLA Wafer Packaging System Revenue in 2 Years	New Product Penetration[2]	Projects With Top 5 Semi Customers[3]

KLA Packaging Revenue including IC Substrate



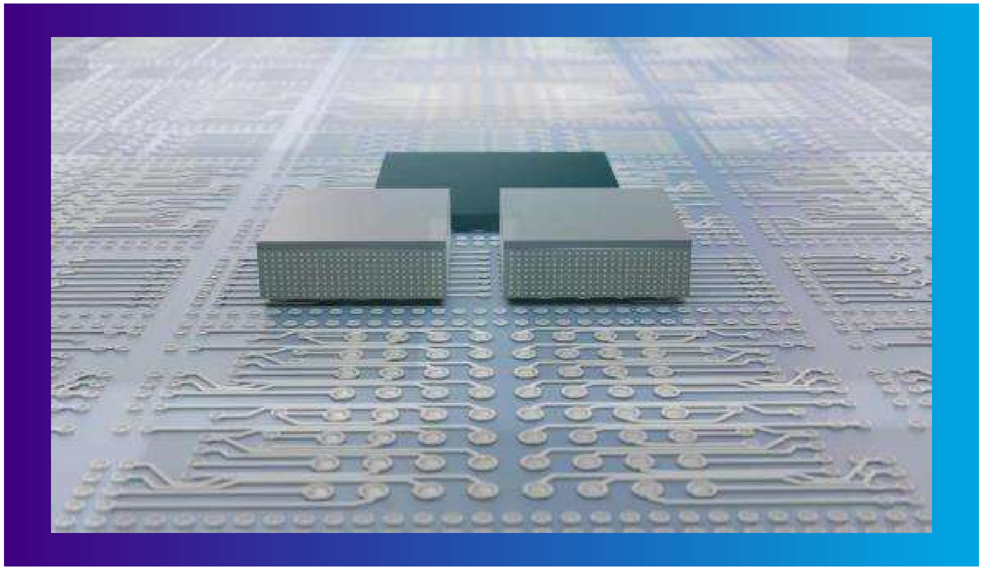
- M&A (ICOS | SPTS | Orbotech → EPC)
- Process Complexity (SPTS → EPC, Semi PC)
- Process Control Intensity (Semi PC)

[1] 2019-24 CAGR (system only, no service)

[2] First product penetration at top 5 semi, top 5 OSATs

[3] Technical engagement on new applications and/or technologies with top 5 semi

Summary : 5 Key Points

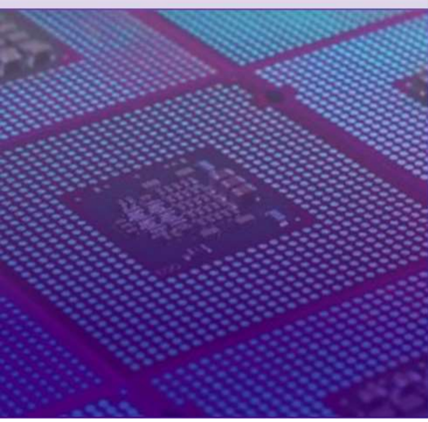


1. The semiconductor technology roadmap is evolving with innovation beyond traditional geometrical scaling
2. Heterogeneous integration is key to enabling overall system performance at lower costs
3. Many packaging technologies are emerging to support higher interconnect density
4. KLA is in a unique position to provide differentiated solutions from front-end to packaging to IC substrates
5. KLA’s advanced packaging revenue continues to grow at a significant pace and is expected to further accelerate in the next 2-3 years

KLA

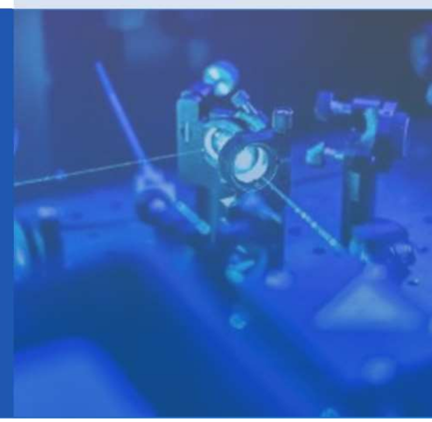
Keep Looking Ahead

Growth



Inflections

R&D



Competition